

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} \quad (1)$ $2\text{MnO}_4^- + 5\text{H}_2\text{C}_2\text{O}_4 + 6\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O} \quad (1)$	1	1		2		
		(ii)	I	31.67 cm ³ accept 31.65 cm ³		1		1	1	1
			II	moles of manganate(VII) = $\frac{31.67 \times 0.0505}{1000} = 1.599 \times 10^{-3} \text{ mol} \quad (1)$ ECF possible from mean volume moles of oxalic acid = $1.599 \times 10^{-3} \times 2.5 = 4.00 \times 10^{-3} \text{ mol} \quad (1)$ ECF possible from equation concentration of saturated solution = $\frac{4.00 \times 10^{-3}}{0.025} \times 10 = 1.60 \text{ mol dm}^{-3} \quad (1)$		3		3	2	